

$$\in \& \cup \cap \Delta - \bar{A} \mid A \mid \subseteq \not\subseteq \subset \not\subset$$

1. Let $X = \{1, 2, 3\}$. Which of the following statements are true?

a) $\{3\} \in X$

F

b) $3 \in X$

T

c) $\{3\} \subseteq X$

T

d) $\{1, 3\} \subseteq X$

T

e) $\{\} \subseteq X$

T

f) $\{\} \in X$

F

g) $\{1, 2, 3\} \subseteq X$

T

h) $\{1, 2, 3\} \subset X$

F

i) $\{1, 2\} \subset X$

T

j) $|X| = 4$

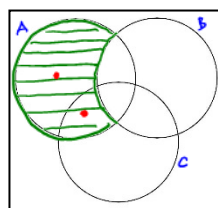
F

k) $2 \subset X$

N/A

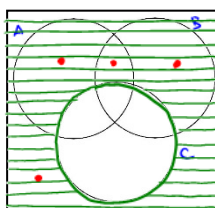
2. Use Venn diagrams to investigate truth or falsity of the following statement:

$$(A - B) \cap \bar{C} = (A \Delta B) - (B \cup C)$$



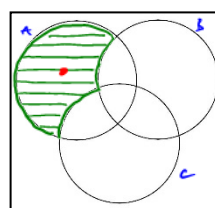
$A - B$

$\cap$



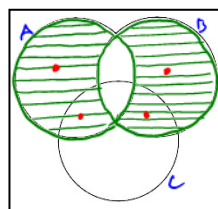
$\bar{C}$

$=$



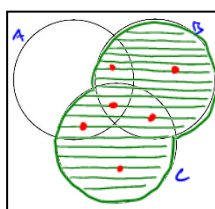
$(A - B) \cap \bar{C}$

TRUE



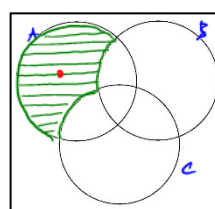
$A \Delta B$

$-$



$B \cup C$

$=$

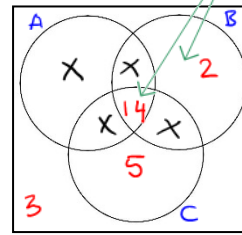


$(A \Delta B) - (B \cup C)$

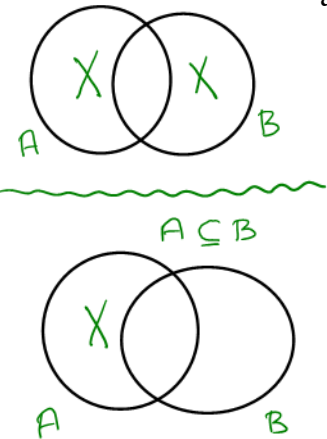
$A = B$ 3.

Consider universe $U = \{1, 2, 3, 4, 5\}$ and give one example of nonempty sets A , B and C that satisfies all of the following:

- $A \cap B = B \cap C$
- $A \subseteq B$
- $|A \cup B| = 3$
- $3 \in \overline{(A \cup B \cup C)}$
- $\{4, 5\} \subset C$



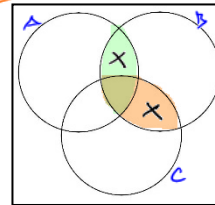
three elem. are here



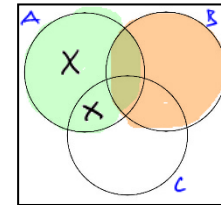
$A = \{1, 4\}$

$B = \{1, 2, 4\}$

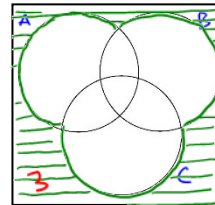
$C = \{1, 4, 5\}$



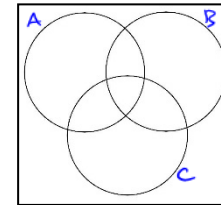
$A \cap B = B \cap C$



$A \subseteq B$

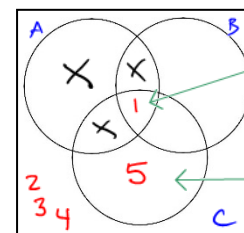


$3 \in \overline{A \cup B \cup C}$



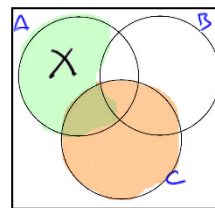
4. Consider universe $U = \{1, 2, 3, 4, 5\}$ and give one example of nonempty sets A , B and C that satisfies all of the following:

- $(A - B) \subseteq C$
- $|C - (A \cup B)| = 1$
- $\{3, 4, 5\} \subset \overline{A}$
- $A \cap C = A \cap B$

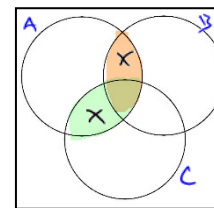


"1" or "2" but not both

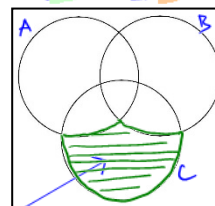
only one el. is here



$(A - B) \subseteq C$



$A \cap C = A \cap B$



$|C - (A \cup B)| = 1$

one el. is here

$A = \{1\}$

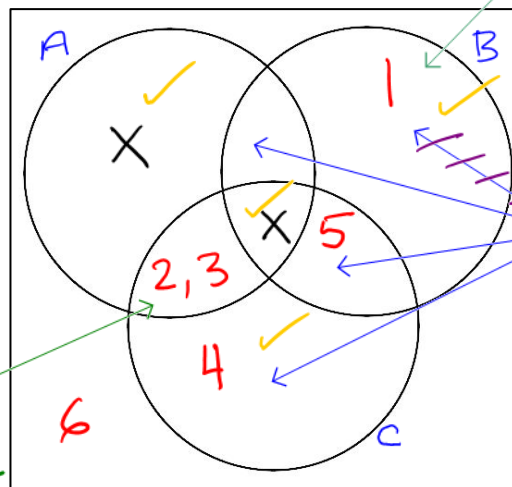
$B = \{1\}$

$C = \{1, 5\}$

5. Provide one example of nonempty sets A, B and C that satisfies all of the following:

$$A \subseteq B$$

- $A \subseteq (B \Delta C)$
- $(\bar{A} \cap B \cap \bar{C}) = \{1\}$
- $((C - B) - \overline{(A \cup B)}) = \{2, 3\}$
- $|((A \cap C) \Delta (B \cup C))| = 3$
- $4 \notin ((A \Delta B) \Delta \bar{C})$



Universe $U = \{1, 2, 3, 4, 5, 6\}$.

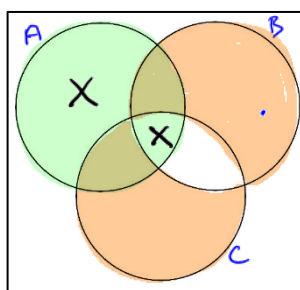
Write your answer:

$$A = \{2, 3\}$$

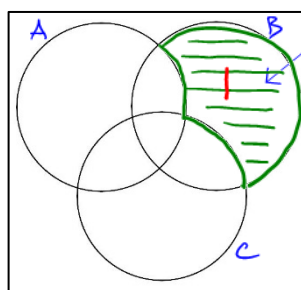
$$B = \{1, 5\}$$

$$C = \{2, 3, 4, 5\}$$

Must write your answer here!
Leaving at least one set empty
implies zero marks.



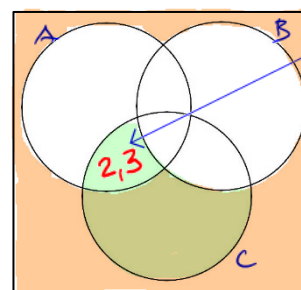
$$A \subseteq (B \Delta C)$$



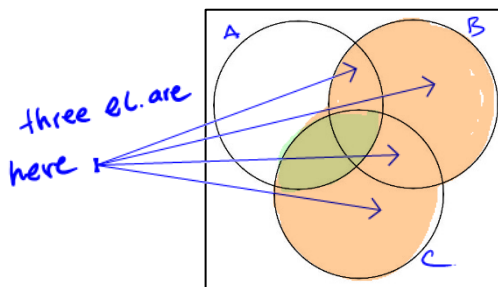
$$\bar{A} \cap B \cap \bar{C} = \{1\}$$

$$1 \in \bar{A} \cap B \cap \bar{C}$$

compare



$$(C - B) - \overline{(A \cup B)} = \{2, 3\}$$



$$|(A \cap C) \Delta (B \cup C)| = 3$$

$$4 \notin ((A \Delta B) \Delta \bar{C})$$

GREEN OR RED, NOT BOTH

$$4 \in \checkmark$$

6. What is the probability that a randomly selected binary string of length 15:

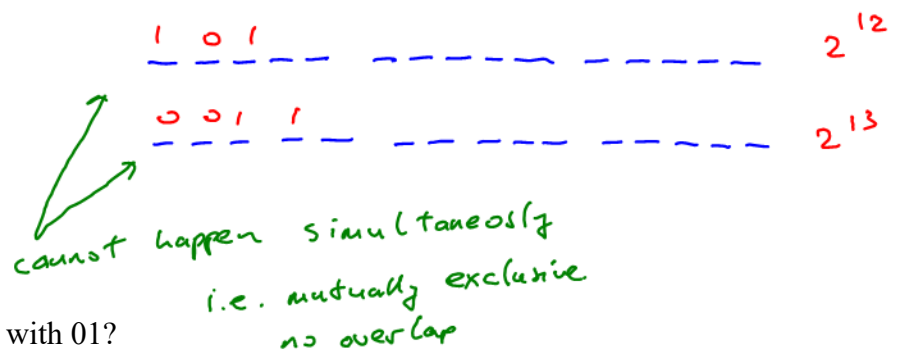
a) start with 101?

$$P_r = \frac{2^{12}}{2^{15}}$$



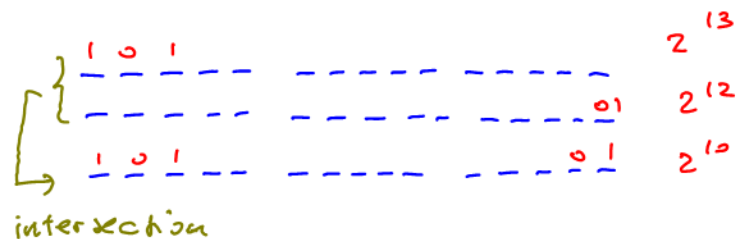
b) start with 101 or start with 0011?

$$P_r = \frac{2^{12} + 2^{11}}{2^{15}}$$



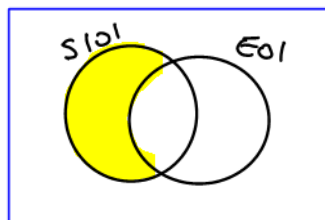
c) start with 101 or end with 01?

$$P_r = \frac{2^{12} + 2^{13} - 2^{10}}{2^{15}}$$



$$|A \cup B| = |A| + |B| - |A \cap B|$$

d) start with 101 but do not end with 01?



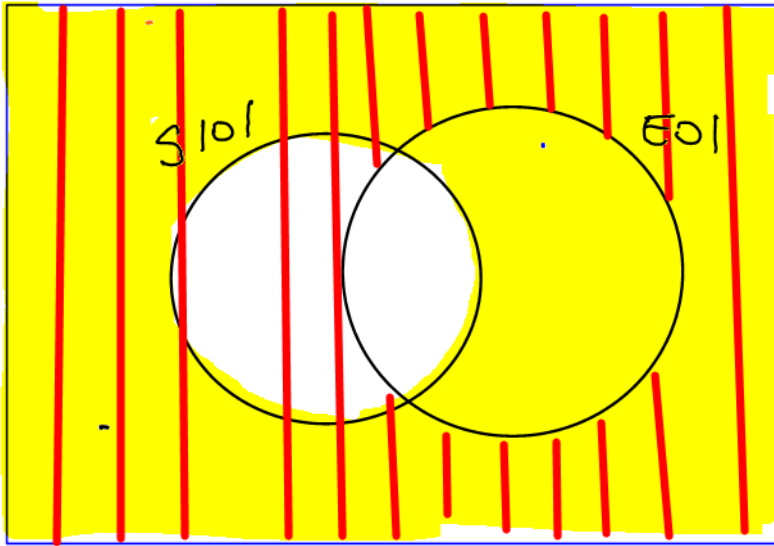
$$|A - B| = |A| - |A \cap B|$$

$$P_r = \frac{2^{12} - 2^{10}}{2^{15}}$$

e) do not start with 101 or do not end with 01?

$\Rightarrow$ yellow or red

$$= \text{all} - |S101 \cap E01|$$



$$Pr = \frac{2^{15} - 2^{10}}{2^{15}}$$

* also NOT S101 OR NOT E01

$$= \text{NOT} (S101 \text{ AND } E01)$$

$$\frac{2^{10}}{2^{15}}$$

$$1 - \frac{2^{10}}{2^{15}}$$

THE SAME

f) do not start with 101 and do not end with 01?

NOT S101 AND NOT E01

$$= \text{NOT} (S101 \text{ OR } E01) = 1 - \frac{2^{13} + 2^{12} - 2^{10}}{2^{15}}$$

part c

$$\frac{2^{13} + 2^{12} - 2^{10}}{2^{15}}$$

7. A student visits an arcade each day after school (Monday to Friday) and plays one game of either Pac-Man, Tetris, or Donkey Kong, but makes sure that each of the three types are played in the first three days of the week.

What is the probability that he plays Pac-Man on consecutive days of a given school week?

Sample Space

$$Pr = \frac{10}{(3)(2)(1)(3)(3)}$$

$\begin{array}{ccccc} M & T & W & R & F \\ \text{P, T, D once} & & & \text{any} & \\ \hline (3) & (2) & (1) & (3) & (3) \end{array}$

for Pac-Man to
be played on
consecutive days
it has to be on

M	T	W	R	F	
P	(2)	(1)	P	P	2
(2)	P	(1)	P	P	2
(2)	(1)	P	P	(3)	6
					t=10